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M. L. DALLA CHIARA

QUANTUM LOGIC AND PHYSICAL MODALITIES

The hypothesis that quantum logic may admit of a modal interpretation seems to be, intuitively, very natural. First of all, as is well known, quantum logic arises essentially in a probabilistic framework, and any probabilistic context gives rise, in a natural way, to some kind of modalities. Further, from a logical point of view, it is very easy to explain the failure of the distributivity laws by means of a suitable *modalization* of the logical connectives.

The search for a modal interpretation of quantum logic gives rise to two different kinds of questions: (1) logical questions: the fundamental problem is to determine a modal calculus *MC* and a translation-function ρ such that for any sentence α of the language of quantum logic, α is a theorem of quantum logic ($\vdash_{QL} \alpha$) iff its modal translation $\rho(\alpha)$ is a theorem of $MC(\vdash_{MC} \rho(\alpha))$; (2) physical questions: what physical meaning may have an introduction of modal concepts in the mathematical formalism of quantum mechanics? Does this logical interpretation represent only a "formal play", or does instead modal logic provide an adequate formal tool for a better clarification of the deepest logical structure of quantum mechanics?

Our first question (the logical one) has been dealt with by Goldblatt [1] and by Dalla Chiara [2], who used essentially a kind of *intensional* semantics for quantum logic, first proposed by Dishkant [3]. Let us call this kind of semantics introduced by Dishkant, *Kripkian semantics* for quantum logic, for it is very close to the semantics that Kripke first proposed for intuitionistic logic and for modal logic. Dishkant's and Goldblatt's Kripkian semantics for quantum logic turn out to be very close to each other. The fundamental difference concerns the accessibility relation (on the Kripkian models) which is reflexive and symmetric in Goldblatt, whereas in Dishkant it satisfies (besides reflexivity) only a weaker condition than symmetry. That is the reason why Goldblatt has proposed as modal calculus *MC* (apt for quantum logic) the *B* calculus, whereas I suggested an intermediate calculus between *T* and *B*. A part of this, the translation function ρ proposed by Goldblatt and by myself turn out to be the same. At present, it seems to me

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that Goldblatt's accessibility relation is to be preferred, for (as we will see) it turns out to be more adequate to what really happens in the mathematical formalism of quantum mechanics. Whereas in Dishkant's paper and in my paper only the case of (sentential) minimal quantum logic (without orthomodular law) was considered, Goldblatt has extended the problem of the modal interpretation also to the case of orthomodular logic.

The aim of my present paper is first of all to develop in a general setting the problem of the modal interpretation of first-order quantum logic (which will be presented in an axiomatic form). The case of minimal quantum logic and of orthomodular logic will be dealt with separately, and the problem of the "modal meaning" of the orthomodular law will be particularly discussed. I will use a general method which will permit me to transform (in a canonical way) any algebraic model of quantum logic into a Kripkian model of quantum logic as well as into a modal model of an extension of the B calculus, and vice versa. These canonical transformations between the models will be used in the last part of the paper in order to try and answer the question on the physical meaning of modal concepts in quantum mechanics.

1. ALGEBRAIC AND KRIPKIAN SEMANTICS FOR FIRST-ORDER MINIMAL QUANTUM LOGIC

As is well known, quantum logic is usually characterized by means of an algebraic semantics. It will be expedient to summarize this standard characterization.

The language L which we refer to, is a standard first-order language (with predicates P^n_m), and the following connectives and quantifiers: $\neg$, $\wedge$, $\rightarrow$, $\forall$. The connectives $\vee$ and $\longleftrightarrow$, and the quantifier $\exists$ are supposed to be defined in the usual way. The symbols $\Rightarrow$, $\forall$, $\mathbb{E}$ will be used as metalogical constants.

DEFINITION 1: An *algebraic orthorealization* of L is a structure $\mathfrak{A} = \langle \mathfrak{B}, v, D \rangle$ where:

(a) $\mathfrak{B}$ is a *complete ortholattice*. By ortholattice we mean a lattice with *maximum*, *minimum*, and *orthocomplement*. In other words it represents a structure $\langle B, \leqslant, \perp, 1, 0 \rangle$, where $\langle B, \leqslant \rangle$ is a lattice, 1 and 0 are respectively the maximum and the minimum of $\mathfrak{B}(\forall x \in B(x \leqslant 1 \text{ and } 0 \leqslant x))$; the operation $\perp$ is an orthocomplement (i.e., $x \sqcup x^{\perp} = 1, x \sqcap x^{\perp} = 0, x = x^{\perp\perp}, x \leqslant y \Rightarrow y^{\perp} \leqslant x^{\perp}$, where $x \sqcup y$ and $x \sqcap y$ represent respectively the *supremum* and the

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infimum of x and y with respect to the lattice-relation $\leqslant$). The *completeness* means that any set X of elements of B has a supremum ($\sqcup X$) and an infimum ($\sqcap X$) in $\mathcal{B}$.

(b) D is a non-empty set (the *universe* of $\mathfrak{A}$). Let L' be an extension of L containing an individual name $\mathbf{d}$ for any $d \in D$.

(c) v realizes the non logical constants of L' and associates to each sentence α of L' a value $v(\alpha)$ in B. For any $\mathbf{d}$, $v(\mathbf{d}) = d$. For any P^n_m, $v(P^n_m)$ is an *n-ary attribute* in $\mathfrak{A}$, that means an element of B^{D^n}. Further, for any sentence α of L', $v(\alpha)$ is determined as follows:

$$v(P^n_m \mathbf{d}_1 \ldots \mathbf{d}_n) = [v(P^n_m)]\,(v(\mathbf{d}_1), \ldots, v(\mathbf{d}_n))$$

$$v(\neg\beta) = v(\beta)^{\perp}$$

$$v(\beta \wedge \gamma) = v(\beta) \sqcap v(\gamma)$$

$$v(\beta \rightarrow \gamma) = \begin{cases} 1 & \text{if } v(\beta) \leqslant v(\gamma) \\ 0 & \text{otherwise} \end{cases}$$

$$v(\forall x \beta) = \sqcap \{v(\beta(x/\mathbf{d}))\}_{d \in D}$$

DEFINITION 2: A sentence α is *true* in $\mathfrak{A}$ ($\models_{\mathfrak{A}} \alpha$) iff $v(\alpha) = 1$. A well formed formula $\alpha(x_1, \ldots, x_n)$ is *true* in $\mathfrak{A}$ ($\models_{\mathfrak{A}} \alpha$) iff $\models_{\mathfrak{A}} \forall x_1 \ldots x_n \alpha$.

DEFINITION 3: A formula α is *algebraically valid* (*A-valid* α) iff for any algebraic orthorealization $\mathfrak{A}$, $\models_{\mathfrak{A}} \alpha$.

It is worth while noticing that our interpretation of the connective $\rightarrow$ corresponds to a kind of *entailment* (or *strong strict implication*). Indeed, as is well known, in the case of minimal quantum logic, the *material conditional* (in the sense of Finch, Mittelstaedt, Hardegree, etc.) is not, in general, reasonably definable.

Now let us describe the Kripkian semantics, which represents a less familiar semantics for quantum logic.

DEFINITION 4: A *Kripkian orthorealization* of L is a structure $\mathcal{K} = \langle U, I, R, \phi, \rhd \rangle$, where:

(a) U is a non-empty set (the *universe* of $\mathcal{K}$). Let L' be an extension of L containing an individual constant $\mathbf{d}$ for each $d \in U$.

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(b) I is a non-empty set, disjoint from U (it represents the set of *worlds* of $\mathcal{K}$).
(c) R is a reflexive and symmetrical relation on I (it represents the *accessibility* relation between the worlds).
(d) ϕ realizes in any world i the non logical constants of L'. For any $i \in I$, $\phi_i(\mathrm{d}) = d$, $\phi_i(P^n_m) \subseteq U^n$.
(e) $\rhd$ is the *forcing-relation*[1] that may hold between worlds and sentences, and satisfies the following conditions:

(e.1)	$i \rhd P^n_m \mathrm{d}_1 \ldots \mathrm{d}_n$	iff $\langle \phi_i(\mathrm{d}_1), \ldots, \phi_i(\mathrm{d}_n)\rangle \in \phi_i(P^n_m)$
(e.2)	$i \rhd \neg\beta$	iff $\forall jRi$:[2] not $j \rhd \beta$.
(e.3)	$i \rhd \beta \wedge \gamma$	iff $i \rhd \beta$ and $i \rhd \gamma$
(e.4)	$i \rhd \beta \rightarrow \gamma$	iff $\forall j$: $j \rhd \beta \Rightarrow j \rhd \gamma$,
(e.5)	$i \rhd \forall x\beta$	iff $i \rhd \beta(\mathrm{d})$ for any $d \in U$.

(f) for any atomic sentence p, $i \rhd p$ iff $\forall jRi\ \exists kRj$: $k \rhd p$.

LEMMA 1: For any sentence α and any i, $i \rhd \alpha$ iff $\forall jRi\ \exists kRj$: $k \rhd \alpha$.

DEFINITION 5: A sentence α is *true* in $\mathcal{K}$ ($\models_{\mathcal{K}} \alpha$) iff for any i, $i \rhd \alpha$. A formula $\alpha(x_1, \ldots, x_n)$ is *true* in $\mathcal{K}$ ($\models_{\mathcal{K}} \alpha$) iff $\models \forall x_1 \ldots x_n \alpha$.

DEFINITION 6: A formula α is *Kripke-valid* (*K-valid* α) iff for any Kripkian orthorealization $\mathcal{K}$, $\models_{\mathcal{K}} \alpha$.

For short we will simply speak of algebraic and of Kripkian realizations of the language L.

As to the problem of an intuitive interpretation of the Kripkian realizations one could try and give a number of imaginative representations of our abstract structures. However, I prefer to postpone this kind of discussion and to deal with this question directly, when we will discuss the problem of the role of this Kripkian semantics in the particular case of quantum mechanics.

Now let $\mathcal{K} = \langle U, I, R, \phi, \rhd\rangle$ be any Kripkian realization of L.

DEFINITION 7: A world i is said to be *orthogonal* to a world j ($i \perp j$) iff not Rij.

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A world i is said to be *orthogonal* to a set of worlds $E \subseteq I$ ($i \perp E$) iff $\forall j \in E$: $i \perp j$.

DEFINITION 8: A set of worlds E is said to be *orthogonally-closed* ($\perp$-closed) iff $\forall i(i \notin E \Rightarrow \exists j(j \perp E$ and $i \not\perp j))$.

One can prove that for any sentence α, the set $I^{\alpha} = [i/i \rhd \alpha]$ is $\perp$-closed. Since I^{α} is usually called (in intensional logics) the *proposition* associated with α, we will call any $\perp$-closed set of worlds a *possible orthoproposition* (or simply an *orthoproposition*) of $\mathcal{K}$.

LEMMA 2:
(a) I and $\emptyset$ are (trivially) $\perp$-closed.
(b) If $\mathcal{F}$ is a set of $\perp$-closed subsets of I, $\cap \mathcal{F}$ is $\perp$-closed.
(c) If $E \subseteq I$ is $\perp$-closed, the set $E^{\textcircled{\perp}} = [i/i \perp E]$ is $\perp$-closed.

THEOREM 1: For any algebraic realization $\mathfrak{A}$ there exists a Kripkian realization $\mathcal{K}^{\mathfrak{A}}$, such that for any sentence α: $\models_{\mathfrak{A}} \alpha$ iff $\models_{\mathcal{K}^{\mathfrak{A}}} \alpha$

Sketch of the proof: Let $\mathfrak{A} = \langle \mathfrak{A}, v, D \rangle$ be an algebraic realization. In order to define $\mathcal{K}^{\mathfrak{A}} = \langle U, I, R, \phi, \rhd \rangle$ we put: $U = D$; $I = B - \{0\}$, Rij iff $i \not\leqslant j^{\perp}$; $\phi_i(\mathrm{d}) = d$; $\phi_i(P^n_m) = [\langle d_1, \ldots, d_n \rangle / i \leqslant v(P^n_m\, \mathrm{d}_1 \ldots \mathrm{d}_n)]$; $i \rhd \alpha$ iff $i \leqslant v(\alpha)$.

THEOREM 2: For any Kripkian realization $\mathcal{K}$ there exists an algebraic realization $\mathfrak{A}^{\mathcal{K}}$, such that for any sentence α: $\models_{\mathcal{K}} \alpha$ iff $\models_{\mathfrak{A}^{\mathcal{K}}} \alpha$.

Sketch of the proof: Let $\mathcal{K} = \langle U, I, R, \phi, \rhd \rangle$ be a Kripkian realization. In order to define $\mathfrak{A}^{\mathcal{K}} = \langle \mathfrak{B}, v, D \rangle$, we put: the domain B of $\mathfrak{B}$ is the set of all orthopropositions of $\mathcal{K}$. If E, F are elements of B: $E^{\perp} = E^{\textcircled{\perp}}$; $E \sqcap F = E \cap F$; $E \cup F = (E^{\perp} \sqcap F^{\perp})^{\perp}$; $1 = I$; $0 = \emptyset$; for any set $\mathcal{F}$ of elements of B, $\sqcap \mathcal{F} = \cap \mathcal{F}$, $\sqcup \mathcal{F} = [\sqcap \{E^{\perp}\}_{E \in \mathcal{F}}]^{\perp}$; $D = U$, $v(\mathrm{d}) = d$; $[v(P^n_m)]\,(d_1, \ldots, d_n) = I^{P^n_m d_1 \ldots d_n}$.

As an immediate consequence of Theorem 1 and Theorem 2 we have: A-valid α iff K-valid α, for any α. In other words, our algebraic and Kripkian semantics characterize the same logic; let us call this logic: (first-order) *minimal quantum logic* (for short MQL).

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2. AN AXIOMATIZATION OF MQL

The MQL calculus is determined by a set of logical axioms plus a set of logical rules. Let us first define the set of all formulas *with an implicational basis* (for short the *I-formulas*) of L.

DEFINITION 9: The set of I-formulas of L is the smallest set Y of formulas s.t.:

(1) Y contains $\alpha \to \beta$ for any α and β of L.

(2) $\neg\alpha$, $\alpha \wedge \beta$, $\forall x\alpha$ belong to Y, if α, β belong to Y.

We will denote any I-formula by $\vec{\alpha}$.

2.1 *Logical axioms of the MQL-calculus*

(1) $\vec{\alpha} \to (\beta \to \vec{\alpha})$

(2) $(\alpha \to (\beta \to \gamma)) \to ((\alpha \to \beta) \to (\alpha \to \gamma))$

(3) $(\alpha \to \beta) \to ((\beta \to \gamma) \to (\alpha \to \gamma))$

(4) $\alpha \to \alpha$

(5) $\alpha \wedge \beta \to \alpha$

(6) $\alpha \wedge \beta \to \beta$

(7) $(\gamma \to \alpha) \to ((\gamma \to \beta) \to (\gamma \to \alpha \wedge \beta))$

(8) $\alpha \wedge \neg\alpha \to \beta$

(9) $(\neg\neg\alpha \to \alpha) \wedge (\alpha \to \neg\neg\alpha)$

(10) $(\alpha \to \beta) \to (\neg\beta \to \neg\alpha)$

(11) $\forall x\alpha(x) \to \alpha(x/t)$

2.2 *Logical Rules of the MQL-calculus*

(i) $$\frac{\alpha, \alpha \to \beta}{\beta}$$

(ii) $$\frac{\alpha}{\beta \to \alpha}$$

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(iii) $\dfrac{\alpha \rightarrow \beta(x)}{\alpha \rightarrow \forall x \beta(x)}$ provided that x is not free in α.

The fundamental difference between MQL and classical logic is determined by axiom (1). In MQL the *a fortiori* principle holds only in the case where α is an I-formula. The failure of the full *a fortiori* axiom forces us to state an *a fortiori* rule (rule (ii)). The concepts of *logical proof* and of *logical theorem* of MQL are defined in the standard way (we will write $\vdash_{\mathrm{MQL}} \alpha$ for "α is a logical theorem of MQL"). Since a deduction theorem cannot be proved for MQL, we define as follows the concept of *formal system formalized in MQL* (for short *MOL-formal system*):

DEFINITION 10: A *MQL-formal system* is a triple $\langle L, C, A \rangle$ where L is a first order language, C is the MQL-calculus (expressed in L), A is a set of sentences of L called the set of *specific axioms* of FS. A formula α is a theorem of FS ($\vdash_{FS} \alpha$) iff there exists a finite number of specific axioms of FS $(\alpha_1, \ldots, \alpha_n)$ s.t. $\vdash_{\mathrm{MQL}} \alpha_1 \wedge \ldots \wedge \alpha_n \rightarrow \alpha$.

A system FS is called *inconsistent* iff a contradiction $(\alpha \wedge \neg\alpha)$ is a theorem of FS; *consistent* otherwise. An *extension* of FS is a formal system FS' whose set of theorems includes all axioms of FS. An algebraic (Kripkian) realization $\mathfrak{A}(\mathcal{K})$ is called a *model* of FS iff all the axioms of FS are true in $\mathfrak{A}(\mathcal{K})$.

THEOREM 3: (Soundness theorem with respect to the algebraic and the Kripkian semantics.) $\vdash_{\mathrm{MQL}} \alpha \Rightarrow A$-valid α (K-valid α).

THEOREM 4: (Completeness theorem with respect to the algebraic and the Kripkian semantics.) A-valid α (K-valid α) $\Rightarrow \vdash_{\mathrm{MQL}} \alpha$.

Let us sketch the proof of the completeness theorem with respect to the Kripkian semantics (owing to Theorems 1 and 2 we obtain, as an immediate consequence, also the completeness with respect to the algebraic semantics). In order to show: not $\vdash_{\mathrm{MQL}} \alpha \Rightarrow \exists \mathcal{K}$ not $\models_{\mathcal{K}} \alpha$, let us use the following lemma (which has been proved in [4]):

LEMMA 3: Any consistent formal system FS admits of a consistent

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extension FS' which is *I-complete* (i.e., $\vdash_{FS'} \vec{\alpha}$ or $\vdash_{FS'} \neg\vec{\alpha}$ for any I-sentence $\vec{\alpha}$) and *I-inductive* (i.e., for any $\vec{\alpha}(x)$, $\vdash_{FS'} \forall x\vec{\alpha}$ whenever for each individual constant $c \vdash_{FS'} \alpha(c)$).

Now let: not $\vdash_{\mathbf{MQL}} \alpha$. Thus the formal system FS whose unique specific axiom is $\neg\alpha$ is consistent. By Lemma 3, there exists a consistent, I-complete and I-inductive extension FS' of FS. In order to define $\mathcal{K} = \langle U, I, R, \phi, \rhd\rangle$ we put: $U =$ the set of all individual constants of FS'; $I =$ the class of all equivalence-classes $[\alpha]_{\equiv}$, where α is a sentence of FS' s.t. not $\vdash_{FS'} \alpha \rightarrow \gamma \wedge \neg\gamma$, and $\alpha \equiv \beta$ iff $\vdash_{FS'} \alpha \longleftrightarrow \beta$; Rij iff not $\vdash_{FS'} \mathrm{i} \rightarrow \neg\mathrm{j}$, where i is a sentence belonging to $i(\mathrm{j} \in j)$; $\phi_i(c) = c$; $\phi_i(P^n_m) = [\langle c_1, \ldots, c_n\rangle / \vdash_{FS'} \mathrm{i} \rightarrow P^n_m c_1 \ldots c_n]$; $i \rhd \beta$ iff $\vdash_{FS'} \mathrm{i} \rightarrow \beta$. One can prove: (1) $\mathcal{K}$ is a Kripkian realization; (2) not $\models_{\mathcal{K}} \alpha$.

3. ORTHOMODULAR LOGIC (QUANTUM LOGIC)

From a syntactical point of view, quantum logic (QL) can be axiomatically characterized as MQL plus an orthomodularity-axiom, for instance the following scheme:

$$(\alpha \rightarrow \beta) \rightarrow (\beta \rightarrow (\alpha \vee (\neg\alpha \wedge \beta))).$$

From a semantical point of view, in the case of the algebraic semantics, we will have, according to the standard characterization:

DEFINITION 11: α is *algebraically quantum valid* iff for any orthomodular algebraic realization $\mathfrak{A}$ of L, $\models_{\mathfrak{A}} \alpha$ (where $\mathfrak{A} = \langle \mathfrak{B}, v, D\rangle$ is *orthomodular* iff $\mathfrak{B}$ is *orthomodular*, i.e., $a \leqslant b \Rightarrow b \leqslant a \sqcup (a^{\perp} \sqcap b)$, for any $a, b \in B$).

What condition corresponds to orthomodularity in the case of our Kripkian semantics? One can answer this question by means of the following definition and theorems:

DEFINITION 12: A Kripkian realization $\mathcal{K}$ is said to be *orthomodular* iff for any orthopropositions E, F of $\mathcal{K}$: $E \cap -F \neq \emptyset \Rightarrow E \cap (E \cap F)^{\Phi} \neq \emptyset$. A sentence α is *Kripke-quantum valid* iff for any orthomodular Kripkian realization $\mathcal{K}$, $\models_{\mathcal{K}} \alpha$.

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THEOREM 5: $\mathfrak{A}$ is orthomodular $\Rightarrow$ $\mathcal{K}^{\mathfrak{A}}$ is orthomodular. $\mathcal{K}$ is orthomodular $\Rightarrow$ $\mathfrak{A}^{\mathcal{K}}$ is orthomodular.

THEOREM 6: $\vdash_{QL} \alpha$ iff α is algebraically quantum valid iff α is Kripke-quantum valid.

4. A MODAL INTERPRETATION OF MQL AND QL

As a consequence of our previous theorems we can now obtain a quite natural modal interpretation of MQL and QL. Let L_1 be a classical modal language, whose non-logical constants are the same as in L. The logical constants of L_1 are the following:

Connectives and quantifiers: $\sim, \&, \supset, \bigwedge$ ($\gamma, \equiv, \bigvee$ defined).

Modal operators: L (necessarily), (M (possible) defined as $\sim L \sim$)

An entailment-connective: $\Longrightarrow$.

DEFINITION 13: A β^B*-Kripkian realization*[3] of L_1 is a structure $\mathcal{K}_1 = \langle U, I, R, \phi, \models\!\!=\rangle$, where $\langle U, I, R, \phi\rangle$ is defined as in Definition 4, and $\models\!\!=$ represents the classical modal relation of *truth*, which may hold between worlds and sentences and satisfies the following conditions:

$\models\!\!=_i P^n_m \mathbf{d}_1 \ldots \mathbf{d}_n$	iff $\langle d_1, \ldots, d_n\rangle \in \phi_i(P^n_m)$
$\models\!\!=_i \sim \beta$	iff not $\models\!\!=_i \beta$
$\models\!\!=_i \beta \,\&\, \gamma$	iff $\models\!\!=_i \beta$ and $\models\!\!=_i \gamma$
$\models\!\!=_i \beta \supset \gamma$	iff $(\models\!\!=_i \beta \Rightarrow \models\!\!=_i \gamma)$
$\models\!\!=_i \bigwedge x\, \beta$	iff $\models\!\!=_i \beta(\mathbf{d})$ for any $d \in D$.
$\models\!\!=_i L\beta$	iff $\forall jRi$: $\models\!\!=_j \beta$
$\models\!\!=_i \beta \Longrightarrow \gamma$	iff $\forall j$: $\models\!\!=_j \beta \Rightarrow \models\!\!=_j \gamma$.

DEFINITION 14: A sentence α is said to be B^B*-valid* iff for any B^B-realization $\mathcal{K}_1$, $\models\!\!=_{\mathcal{K}_1} \alpha$ (i.e., for any world i of $\mathcal{K}_1$, $\models\!\!=_i \alpha$).

The B^B logic admits of an axiomatization s.t.: $\vdash_{B^B} \alpha$ iff B^B-valid α. Now

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let us define the following translation ρ of the ortholanguage L in the classical language L_1:

$$\rho(p) = LMp \text{ for any atomic formula } p \text{ of } L$$

$$\rho(\neg\alpha) = L \sim \rho(\alpha)$$

$$\rho(\alpha \wedge \beta) = \rho(\alpha) \,\&\, \rho(\beta)$$

$$\rho(\alpha \rightarrow \beta) = \rho(\alpha) \Longrightarrow \rho(\beta)$$

$$\rho(\forall x\alpha) = \bigwedge x \rho(\alpha)$$

One can prove:

THEOREM 6: Any Kripkian orthorealization $\mathcal{K}$ of L can be transformed in a B^B-modal realization $\mathcal{K}_1$ of L_1, and conversely any B^B-realization $\mathcal{K}_1$ can be transformed in a Kripkian orthorealization $\mathcal{K}$ s.t. for any world i: $i \rhd \alpha$ iff $\underset{i}{\models} \rho(\alpha)$, for any sentence α of L.

THEOREM 7: For any α of L:

$\underset{\mathbf{MQL}}{\vdash} \alpha$ iff $\underset{B^B}{\vdash} \rho(\alpha)$

$\underset{\mathbf{QL}}{\vdash} \alpha$ iff $\underset{B^B + \rho(\mathrm{Ort})}{\vdash} \rho(\alpha)$, where $\rho(\mathrm{Ort})$ represents the modal translation of the orthomodular axiom.

An interesting question concerns the modal meaning of the orthomodular law: what is asserted, intuitively, by $\rho(\mathrm{Ort})$? In order to discuss this problem, let us call an *orthosentence* of L_1 any sentence of L_1 which is the modal translation of a sentence of L (for instance LMp is an orthosentence, but $\sim p$ is not an orthosentence). One can show that the ρ-translation of the orthomodular law is equivalent (in any B^B-realization) to the following schema: $\alpha \,\&\, \sim \beta \supset M(\alpha \,\&\, L \sim (\alpha \,\&\, \beta))$, where α and β represent orthosentences. Intuitively, we could read this modal version of the orthomodular law as follows: if in the actual world a first event does happen and a second event does not happen, then there is a possible world where the first event still happens, whereas the first and the second event are necessarily incompatible (for instance: if in the actual world it rains and it is not midnight, then there is a possible world where it rains, however it cannot rain at

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midnight[4]). Of course, such a postulate represents a somewhat strong modal requirement which, intuitively, seems to be not very perspicuous. Roughly speaking, we could say that an orthomodular B^B-realization is a realization which has "sufficiently many possible worlds", such that any *contingent* incompatibility (between orthosentences) *might be essential*.

5. MODAL CONCEPTS IN THE MATHEMATICAL FORMALISM OF QUANTUM MECHANICS

What intuitive meaning may have this kind of intensional and of modal semantics in the particular case of quantum mechanics? In [1] Goldblatt observed that "whether or not a description [associated with the Kripkian semantics] can be found that has a significance for quantum theory remains to be seen." Let us try and give some answer to this question.

As is well known, in the case of quantum mechanics, one can refer to a class of *natural* algebraic orthorealizations of a sublanguage of the theory. Such sublanguage L is determined as follows: the atomic w.f.f. of L have the form $q_i \in \mathbf{b}, q_i \in x$, where q_i is a special variable ranging over the *ideal* possible values of an *observable* Q_i,[5] $\mathbf{b}$ is the name for a Borel set b of real numbers, x is a variable ranging over all Borel subsets of the real line (we will write also $P_i\mathbf{b}$ and $P_i x$ for $q_i \in \mathbf{b}$ and $q_i \in x$). The molecular w.f.f. are generated by means of our orthoconnectives and quantifiers.

Now, if s is a physical system and H is a Hilbert space associated with s (according to the standard rules of quantum mechanics), one can define, in a natural way, an algebraic realization $\mathfrak{A}^H = \langle \mathfrak{B}^H, v, D\rangle$ of the language L. $\mathfrak{B}^H$ is the complete ortholattice of all closed subspaces of H; D is the set of all Borel subsets of the real line; $v(\mathbf{b}) = b$; $v(P_i)(b) = [\psi/\mathrm{prob}^{Q_i}[\psi, b] = 1]$, where $\mathrm{prob}^{Q_i}[\psi, b]$ (the probability that a value for Q_i in the state ψ can be found in b) is determined by the standard rules of quantum mechanics.

Let us state:

DEFINITION 15: $\underset{\psi}{\models} \alpha$ (α is *true* in the physical state ψ) iff $\psi \in v(\alpha)$.

We want now to study how concretely behave our Kripkian orthorealization $\mathcal{K}^{\mathfrak{A}^H}$ and our modal realization $\mathcal{K}_1^{\mathfrak{A}^H}$, which (owing to Theorem 1 and Theorem 6) are associated with the algebraic realization $\mathfrak{A}^H$. As a consequence of our previous theorems, we will obtain:

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$$\mathcal{K}^{\mathfrak{A}H} = \langle D, I, R, \phi, \rhd \rangle; \quad \mathcal{K}_1^{\mathfrak{A}H} = \langle D, I, R, \phi, \models \rangle$$

where D is the set of all Borel sets of real numbers; I is the set of all non-null closed subspaces X of H; R is the non-orthogonality relation in $\mathfrak{A}^H$; $\phi_X(\mathbf{b}) = b; \phi_X(P_m) = [b/X \subseteq v(P_m \mathbf{b})]$; $X \rhd \alpha$ iff $X \subseteq v(\alpha)$; $\underset{X}{\models} p$ iff $X \rhd p$, for any atomic sentence of the modal language L_1 associated with L.

From an intuitive point of view, one can say that any ψ represents, as is usually maintained, a *possible physical state*, any subspace X (which is in $\mathcal{K}^{\mathfrak{A}H}$ and in $\mathcal{K}_1^{\mathfrak{A}H}$) represents a *physical world* which may be described by an orthosentence α of L such that $X = v(\alpha)$; any orthoproposition E of $\mathcal{K}^{\mathfrak{A}H}$ represents a *set of physical worlds*, which may be described by an orthosentence α such that $I^\alpha = E$.

Among the physical worlds it is interesting to consider the *unidimensional* physical worlds X^ψ, which contain only one normalized vector ψ. As a consequence of our previous results we will have: $X^\psi \rhd \alpha$ iff $X^\psi \subseteq v(\alpha)$ iff $\psi \in v(\alpha)$ iff $\underset{\psi}{\models} \alpha$ iff $\underset{X\psi}{\models} \rho(\alpha)$. (we will write also $\underset{\psi}{\models} \rho(\alpha)$ instead of $\underset{X\psi}{\models} \rho(\alpha)$).

What is the physical meaning of the modal operators in $\mathcal{K}_1^{\mathfrak{A}H}$? In order to answer this question, let us first recall that (according to the standard rules of quantum mechanics) one can associate to any vector ψ a (non-classical) measure function μ_ψ which associates a probability value with any orthosentence of L. On this ground one can easily calculate the probabilistic meaning of the following modal prefixes (when applied to the modal translations of sentences of L):

$\underset{\psi}{\models} M\rho(\alpha)$ iff $\mu_\psi(\alpha) \neq 0$ (the possibility of the modal translation of α is true in a physical state ψ iff α has in ψ probability different from 0).

$\underset{\psi}{\models} L\ \rho(\alpha)$ iff $\mu_\phi(\alpha) = 1$, for any ϕ (that means α represents the *certain* sentence of the language).

$\underset{\psi}{\models} LM\ \rho(\alpha)$ iff $\mu_\psi(\alpha) = 1$ (the necessity of the possibility of the modal translation of α is true in ψ iff α has in ψ probability 1).

$\underset{\psi}{\models} LL\ \rho(\alpha)$ iff $\underset{\psi}{\models} L\ \rho(\alpha)$

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$\underset{\psi}{\models} \Pi L\, \rho(\alpha)$ iff $\underset{\psi}{\models} L\, \rho(\alpha)$, where Π represents any modal prefix.

Whereas LL reduces to L, MM does not reduce to M. In other words, we have a kind of "S_4-reduction of modalities" only for L. This asymmetrical behaviour between the necessity and the possibility operators is possible, since the modal relations stated above hold *only* for modal translations of sentences of L (and not for any sentence of L_1) (thus one cannot prove here: $[L\, \rho(\alpha) \supset LL\, \rho(\alpha)] \supset [MM\, \rho(\alpha) \supset M\, \rho(\alpha)]$). In this way, we have obtained a modal framework, where the modal prefixes, when applied to orthosentences of L_1, receive a quite natural probabilistic interpretation.

As a conclusion, let us ask whether these model-theoretical results may have some bearing on the general philosophy of quantum mechanics. It seems to me that all this can lead, in a natural way, to the following claim: from a logical point of view quantum logic is not really *essential* to the logical development of quantum mechanics, since the role played by this logic can be equivalently substituted by a form of classical modal logic. Perhaps this conclusion can give some psychological help to those people who think that a theoretical situation essentially founded on a *mixture of logics*, is very disagreeable. And without any doubt, if quantum logic were essential to the logical development of quantum mechanics, quantum mechanics as a whole would be founded on a mixture of logics: for, so far, it is quite improbable that we will formalize the whole of mathematics needed for quantum mechanics in a quantum logical calculus such as QL. As a logician, I must say that I do not feel any particular allergy to a situation of mixture of logics. On the contrary, it seems to me that, generally, a form of plurality of logics cannot be avoided in modern science. In any case, even if philosophically we do not trust a unique priviliged logic, from a logical point of view any reduction of a logic to an other logic represents a relevant result. However, in this particular case, concerning quantum logic, can we really assert that we have completely reduced quantum logic to classical logic? As we have seen, we have only proved the following metatheorems: $\underset{\mathrm{MQL}}{\vdash} \alpha$ iff $\underset{B^B}{\vdash} \rho(\alpha)$ and $\underset{\mathrm{QL}}{\vdash} \alpha$ iff $\underset{B^B + \rho(\mathrm{Ort})}{\vdash} \rho(\alpha)$. But B^B and $B^B + \rho(\mathrm{Ort})$ are not *simply* classical logic. Indeed they represent two very particular extensions of classical logic. One could ask: why just the modal system B^B fits with quantum mechanics? And why this curious orthomodular postulate, whose modal version does not really sound as a law, which must be, classically, *true*

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in all possible worlds? Thus, the general question concerning the disagreeable mixture of logics in quantum mechanics seems to remain still open.

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NOTES

[1] Intuitively, one can read $i \rhd \alpha$ as "α holds in the world i". As is customary, we prefer to speak of "forcing" and not simply of "truth", because, unlike in Kripkian modal semantics, here the Kripkian worlds do not generally satisfy the strong *tertium non datur*-property (in other words, we do not have: for any α, $i \rhd \alpha$ or $i \rhd \neg\alpha$).

[2] We read $\forall jRi$ as $\forall j(Rij \Rightarrow \ldots)$ and $\exists jRi$ as $\exists j(Rij$ and $\ldots)$. Of course, for the symmetry of R, $\forall iRj$ and $\forall jRi$ ($\exists iRj$ and $\exists jRi$) are equivalent.

[3] The expression B^B means "the B modal system + the so called *Barcan-formula*". See [6], pp. 57–59; 141–174.

[4] For the sake of simplicity, I am supposing that "it rains" and "it is midnight" are equivalent to some orthosentences. This is not a too strange hypothesis, for, just in the case of quantum mechanics we we will obtain that any atomic sentence p turns out to be equivalent (in the modal realization) to LMp.

[5] For a systematic description of this language see [5].

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